EIGENVALUE CONVERGENCE FOR POLYNOMIALS IN INDEPENDENT GUE MATRICES

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Abstract. We prove that the empirical eigenvalue distribution of any fixed scalar polynomial in independent normalized GUE matrices converges weakly almost surely to the Brown measure of the same polynomial in free semicircular elements. The result holds for every coupling across matrix sizes. We also establish almost-sure and mean convergence of normalized logarithmic determinants for affine matrix pencils in the GUE inputs.

1. Introduction

Let $d \geq 1$ and let $H^{(N)} := (H_1^{(N)}, \dots, H_d^{(N)})$ consist of independent $N \times N$ matrices from the Gaussian unitary ensemble (GUE), each with density proportional to $\exp(-N \operatorname{Tr}(H^2)/2)$ on the real vector space of Hermitian matrices. The tuples for different N are defined on a common probability space with arbitrary coupling across sizes. Here Tr denotes the unnormalized matrix trace and $\operatorname{tr}_N := N^{-1} \operatorname{Tr}$. Let $s := (s_1, \dots, s_d)$ be a free centered semicircular family of variance one in a finite von Neumann algebra $\mathcal{M}$ with faithful normal tracial state τ. Each s_j has spectral density $(2\pi)^{-1}\sqrt{4 - x^2}$ on $[-2, 2]$. Freeness means that the trace of a product of centered elements from the unital algebras generated by the individual s_j vanishes whenever consecutive factors come from distinct algebras [NS06, Definition 2.16, Remark 2.17(2), and Definition 5.3].

For each scalar polynomial Q in the input variables, $\operatorname{tr}_N(Q(H^{(N)}))$ converges in probability to $\tau(Q(s))$. The maximum of the input operator norms converges in probability to 2, the right endpoint of the variance-one semicircle law [CGP22, pp. 1–2 of arXiv v1]. For a self-adjoint scalar polynomial P, this gives convergence of the empirical eigenvalue distribution of $P(H^{(N)})$ to the spectral distribution of $P(s)$. For a general polynomial, the eigenvalues can lie in the complex plane, and the corresponding limit is given by the Brown measure.

We use the Brown measure convention of [MS17, Chapter 11, Theorem 5, Definition 6 and pp. 268–270].

Definition 1.1. Let $\mathcal{M}$ be a finite von Neumann algebra with faithful normal tracial state τ, and let $a \in \mathcal{M}$. The Brown measure of a is the measure μ_a^{Br} given by

$$\mu_a^{\mathrm{Br}} := \frac{1}{2\pi} \Delta u_a, \qquad u_a(z) := \tau(\log |a - z 1_{\mathcal{M}}|).$$

Here $|b| := (b^* b)^{1/2}$, the trace of the logarithm is the spectral integral with respect to τ, and Δ acts in the sense of distributions on $\mathbb{C}$.

The potential u_a is locally integrable, and μ_a^{Br} is a probability measure supported in the spectrum of a. Moreover, $u_a(z) = \int_{\mathbb{C}} \log |z - w| \, d\mu_a^{\mathrm{Br}}(w)$, and for normal a the Brown measure agrees with its spectral distribution.

Belinschi, Śniady and Speicher [BŚS18, Section 7] formulate the eigenvalue convergence conjecture for general polynomials in independent self-adjoint Gaussian matrices. Han [Han26, Theorem 1.2] proves weak convergence in probability for polynomials in independent complex Ginibre matrices using only the unstarred variables, with the corresponding free circular limit. Two independent GUE inputs can be combined as $(H_1^{(N)} + \mathrm{i} H_2^{(N)})/\sqrt{2}$ to produce a normalized

complex Ginibre matrix, whose real and imaginary entry coordinates are independent centered Gaussians of variance $1/(2N)$. Writing a general GUE polynomial in these variables introduces their adjoints, a case discussed in [Han26, Question 1.8]. We prove almost-sure weak convergence for the full class of GUE polynomials.

Theorem 1.2. *Let $d \geq 1$ and fix $P \in \mathbb{C}\langle x_1, \ldots, x_d\rangle$, where $x_1, \ldots, x_d$ are self-adjoint indeterminates. Let $H^{(N)}$ be as above, and let $s_1, \ldots, s_d \in \mathcal{M}$ be freely independent centered semicircular elements of variance one in a finite von Neumann algebra with faithful normal tracial state τ. Then the empirical measure*

$$\mu_{P(H^{(N)})} := \frac{1}{N}\sum_{k=1}^{N} \delta_{\lambda_k(P(H^{(N)}))}$$

converges weakly almost surely to $\mu^{\mathrm{Br}}_{P(s)}$ for every coupling across sizes. The eigenvalues are counted with algebraic multiplicity. For each fixed P and each coupling, the convergence holds on one event of probability one against every bounded continuous real-valued test function on $\mathbb{C}$.

We first state the estimate for a matrix pencil with identity constant term. For a Hermitian tuple $H = (H_1, \ldots, H_d)$ of size N, let

$$T_N(H) := I_m \otimes I_N + \sum_{j=1}^{d} R_j \otimes H_j, \tag{1.1}$$

where the coefficient matrices $R_1, \ldots, R_d \in M_m(\mathbb{C})$ are fixed. We call such a pencil monic. For a square matrix B, write $s_{\min}(B) := \min_{\|v\|=1} \|Bv\|$.

Theorem 1.3. *Let $d, m \geq 1$, and let $R_1, \ldots, R_d \in M_m(\mathbb{C})$ be arbitrary fixed matrices. There exists $A > 0$ such that*

$$\sum_{N=1}^{\infty} \mathbb{P}\{s_{\min}(T_N(H^{(N)})) < N^{-A}\} < \infty. \tag{1.2}$$

For each $M > 0$ there exist $C, \alpha, \varepsilon_0 > 0$, independent of N, such that for $0 < \varepsilon \leq \varepsilon_0$,

$$\begin{aligned} &\mathbb{P}\{s_{\min}(T_N(H^{(N)})) \leq \varepsilon,\ \max_j \|H_j^{(N)}\| \leq M\} \\ &\qquad \leq CN^{5/2}\varepsilon^{\alpha} \exp(CN^{5/2}\varepsilon^{\alpha}). \end{aligned} \tag{1.3}$$

For arbitrary fixed $A_0, \ldots, A_d \in M_m(\mathbb{C})$, write

$$L_N(H) := A_0 \otimes I_N + \sum_{j=1}^{d} A_j \otimes H_j, \qquad L := A_0 \otimes 1_{\mathcal{M}} + \sum_{j=1}^{d} A_j \otimes s_j. \tag{1.4}$$

We use $L_N := L_N(H^{(N)})$ for the random evaluation and $\tau_m := \mathrm{tr}_m \otimes \tau$ for the normalized trace on $M_m(\mathcal{M})$. For $b \in M_m(\mathcal{M})$ the Fuglede–Kadison determinant is $\Delta_{\tau_m}(b) := \exp(\tau_m(\log|b|))$, with $\log 0 = -\infty$ and $\exp(-\infty) = 0$ [MS24, Definition 2.1(ii) and Remark 2.2(i)]. For a complex matrix B of size k, write $\Delta_k(B) := |\det B|^{1/k}$ for the determinant with respect to the normalized trace [MS24, Proposition 2.3(vi)].

Theorem 1.4. *Let L_N and L be as in (1.4), and put*

$$U_N := \frac{1}{mN}\log|\det L_N|, \qquad u := \tau_m(\log|L|).$$

For every coupling across sizes, $U_N \to u$ and $\Delta_{mN}(L_N) \to \Delta_{\tau_m}(L)$ almost surely. For every $0 < p < \infty$,

$$\mathbb{E}|\Delta_{mN}(L_N) - \Delta_{\tau_m}(L)|^p \longrightarrow 0.$$

If $u > -\infty$, then U_N is integrable for all sufficiently large N, and $\mathbb{E}|U_N - u| \to 0$, in particular $\mathbb{E}U_N \to u$. In this case there are $b, C, \varepsilon_0 > 0$ and N_0 such that

$$\mathbb{P}\{s_{\min}(L_N) \le \varepsilon\} \le CN^{5/2}\varepsilon^b \qquad (N \ge N_0,\ 0 < \varepsilon \le \varepsilon_0,\ N^{5/2}\varepsilon^b \le 1). \tag{1.5}$$

If $u = -\infty$, then U_N^+ is integrable for every N and $\mathbb{E}U_N \to -\infty$ in the extended real sense.

The first step is to show that a small singular value puts the Hermitian input tuple close to a tuple for which the pencil is singular. In Section 2, we compress an approximate kernel vector and the action of the inputs on its components to a space of dimension bounded in terms of m and d. A polynomial distance inequality in that dimension yields a nearby exact kernel equation, which we lift to the original space. Section 3 then bounds the Gaussian measure of a neighborhood of the zero set of the determinant. In Section 4, a rank argument ensures that the affine determinant is a nonzero polynomial for all large sizes when u is finite. We then let the norm cutoff depend on the singular-value threshold to obtain (1.5). Gaussian concentration for truncated logarithms yields the almost-sure determinant limit.

In Section 5, a Schur complement identifies the scalar logarithmic potential with the normalized logarithmic determinant of an affine pencil, up to the fixed pencil size. Hermitization and logarithmic truncation identify the limit, following the method of [Han26, Section 5]. Fubini's theorem and local square-integrability of the logarithmic kernel give pathwise local L^1 convergence of the potentials and hence weak convergence of the eigenvalue measures.

2. Distance to singular pencils

Proposition 2.1. *Fix $A_0, \dots, A_d \in M_m(\mathbb{C})$ and $M > 0$. For a tuple $H = (H_1, \dots, H_d)$ of Hermitian $N \times N$ matrices, use the pencil $L_N(H)$ from (1.4). There exist $C, \alpha, \varepsilon_0 > 0$ such that, for every N and every such H with $\max_j \|H_j\| \le M$, the following holds. If $s_{\min}(L_N(H)) \le \varepsilon \le \varepsilon_0$ with $\varepsilon > 0$, there are Hermitian $N \times N$ matrices $\Delta H_1, \dots, \Delta H_d$ such that $L_N(H + \Delta H)$ is singular and*

$$\left(\sum_{j=1}^{d} \|\Delta H_j\|_{\mathrm{HS}}^2\right)^{1/2} \le C\varepsilon^{\alpha}. \tag{2.1}$$

Here $\|X\|_{\mathrm{HS}} := (\mathrm{Tr}(X^*X))^{1/2}$ is the Hilbert–Schmidt norm, with the unnormalized trace on the appropriate matrix size.

Proof. Choose a unit vector $v = (v_1, \dots, v_m) \in \mathbb{C}^m \otimes \mathbb{C}^N$ with $\|L_N(H)v\| \le \varepsilon$. Let $W := \mathrm{span}\{v_1, \dots, v_m\}$, and let $W_1 := \sum_{j=1}^d P_{W^\perp} H_j W$, where $P_{W^\perp}$ is the orthogonal projection onto $W^\perp$. Writing $k := \dim W$ and $\ell := \dim W_1$, we have $1 \le k \le m$ and $0 \le \ell \le dk$. Choose isometries $U : \mathbb{C}^k \to \mathbb{C}^N$ and $V : \mathbb{C}^\ell \to \mathbb{C}^N$ onto W and W_1, respectively, with $\mathbb{C}^0 = \{0\}$. For each j put $C_j := U^*H_jU$ and $B_j := V^*H_jU$, so that C_j is Hermitian and $B_j \in M_{\ell,k}(\mathbb{C})$. By the definition of W_1,

$$H_jU = UC_j + VB_j. \tag{2.2}$$

Write $w := (U^*v_1, \dots, U^*v_m) \in \mathbb{C}^m \otimes \mathbb{C}^k$. The constant term preserves $\mathbb{C}^m \otimes W$. The two coordinate vectors of $L_N(H)v$ in the orthogonal summands $\mathbb{C}^m \otimes W$ and $\mathbb{C}^m \otimes W_1$ are

$$(A_0 \otimes I_k)w + \sum_{j=1}^{d} (A_j \otimes C_j)w \quad \text{and} \quad \sum_{j=1}^{d} (A_j \otimes B_j)w. \tag{2.3}$$

Equation (2.2) gives no component outside these summands, so the sum of the squared norms in (2.3) is $\|L_N(H)v\|^2$.

Use Hilbert–Schmidt orthonormal real coordinates for each C_j, and split the entries of each B_j and of w into real and imaginary parts. For each (k, ℓ) in the indicated ranges, these choices

give $p_{k,\ell} := dk^2 + 2d\ell k + 2mk$ real coordinates for $x := (C, B, w)$. Let $F_{k,\ell} : \mathbb{R}^{p_{k,\ell}} \to \mathbb{R}^{2m(k+\ell)+1}$ be the polynomial map whose coordinates are the real and imaginary parts of (2.3) and the scalar $\|w\|^2 - 1$. The map is quadratic and depends only on the coefficients $A_0, \dots, A_d$ and the pair (k, ℓ). Since $\|w\| = 1$ and $\|C_j\|_{\mathrm{HS}}, \|B_j\|_{\mathrm{HS}} \le \sqrt{k}M$, the Euclidean norms of x and its image satisfy

$$|x|^2 \le 2dkM^2 + 1, \qquad |F_{k,\ell}(x)| \le \varepsilon. \tag{2.4}$$

The real polynomial distance inequality of Kurdyka and Spodzieja, as stated in [KSS16, p. 2, (KS2)], gives the following bound for a polynomial map $F : \mathbb{R}^p \to \mathbb{R}^q$ of degree $D \ge 1$ with a nonempty zero set. For some $c_F > 0$ and every $y \in \mathbb{R}^p$,

$$|F(y)| \ge c_F \left(\frac{\operatorname{dist}(y, F^{-1}(0))}{1 + |y|^2} \right)^{D(6D-3)^{p-1}}.$$

When $F_{k,\ell}$ has a zero, its zero set is closed, so the distance from x is attained at some $x' := (C', B', w')$. Using $D = 2$ and (2.4) gives

$$|x' - x| \le C_{k,\ell} \varepsilon^{\alpha_{k,\ell}}, \qquad \alpha_{k,\ell} := (2 \cdot 9^{p_{k,\ell}-1})^{-1}, \tag{2.5}$$

where $C_{k,\ell}$ depends only on the map and M. If the zero set is empty, then $|F_{k,\ell}|$ has a positive minimum on the closed ball $|x|^2 \le 2dkM^2 + 1$. There are only finitely many pairs (k, ℓ), so we can choose $\varepsilon_0 \le 1$ below these positive minima, and choose a common exponent $\alpha > 0$ and a constant C_1 in (2.5).

With the original U and V, define

$$\Delta H_j := U(C_j' - C_j)U^* + V(B_j' - B_j)U^* + U(B_j' - B_j)^* V^*.$$

This matrix is Hermitian and satisfies $(H_j + \Delta H_j)U = UC_j' + VB_j'$ by (2.2) and $V^*U = 0$. The coordinates of $L_N(H + \Delta H)(I_m \otimes U)w'$ are therefore the expressions in (2.3) with C_j, B_j, w replaced by C_j', B_j', w'. Both expressions vanish because $F_{k,\ell}(x') = 0$, and its normalization coordinate gives $\|(I_m \otimes U)w'\| = 1$. This produces a kernel vector even if its components span a proper subspace of W. The three blocks in ΔH_j are pairwise Hilbert–Schmidt orthogonal, so

$$\sum_{j=1}^{d} \|\Delta H_j\|_{\mathrm{HS}}^2 = \sum_{j=1}^{d} \left(\|C_j' - C_j\|_{\mathrm{HS}}^2 + 2\|B_j' - B_j\|_{\mathrm{HS}}^2 \right) \le 2|x' - x|^2.$$

This proves (2.1) with $C = \sqrt{2}C_1$. □

3. Gaussian tubes and the least singular value

Proposition 2.1 reduces the small singular value event to proximity to a real polynomial zero set. We first bound the Gaussian measure of the points within a given distance of such a set.

Lemma 3.1. *Let q be a nonzero, nonnegative real polynomial on $\mathbb{R}^n$ of degree at most D, where $D \ge 1$, and suppose $Z := q^{-1}(0)$ is nonempty. If g is a standard Gaussian vector in $\mathbb{R}^n$, then for every $\rho > 0$,*

$$\mathbb{P}\{\operatorname{dist}(g, Z) \le \rho\} \le CD\sqrt{n}\,\rho \exp\bigl(C(D+1)\sqrt{n}\,\rho\bigr)$$

for a universal constant $C > 0$.

Proof. Let X be uniform in the ball of radius $\sigma > 0$ centered at the origin in $\mathbb{R}^n$. For a smooth hypersurface Y given by a regular level of a polynomial of degree at most D, the codimension-one case of Lotz's tube estimate [Lot15, Theorem 1.1] gives

$$\mathbb{P}\{\operatorname{dist}(X, Y) \le \rho\} \le 4 \sum_{r=1}^{n} \binom{n}{r} (2D\rho/\sigma)^r (1 + \rho/\sigma)^{n-r}.$$

The binomial formula rewrites the bound as

$$4\left[\left(1+\frac{(2D+1)\rho}{\sigma}\right)^n-\left(1+\frac{\rho}{\sigma}\right)^n\right]. \tag{3.1}$$

To apply this bound to Z, fix $\eta>0$ and consider the zeros of q in the closed ball of radius $\sigma+\rho$. Since Z is closed, every point of the tube in the ball of radius σ has a nearest zero in this larger ball. If there are no such zeros, the tube event in the ball of radius σ is empty. Otherwise, near each such zero y there is a point p_y with $|p_y-y|<\eta/2$ and $q(p_y)>0$, since a nonzero nonnegative polynomial cannot vanish on an open ball. By compactness, finitely many open balls of radius $\eta/2$ centered at these zeros cover the relevant zero set. Choose a positive regular value δ of q less than all the values $q(p_y)$ for the chosen centers, using Sard's theorem. For each relevant zero y', the segment from y' to the positive point associated with a covering ball lies within distance η of y' and meets $q^{-1}(\delta)$. Every point in the ball of radius σ within distance ρ of Z is therefore within distance $\rho+\eta$ of $q^{-1}(\delta)$. The latter is a smooth hypersurface defined by a polynomial of degree at most D. Applying (3.1) with $\rho+\eta$ and letting η decrease to zero gives (3.1) for Z.

For the Gaussian estimate, first sample $S>0$ with the chi distribution with $n+2$ degrees of freedom and then sample a point uniformly in the ball of radius S. With ω_n the volume of the unit ball in $\mathbb{R}^n$, the resulting density at $x\in\mathbb{R}^n$ is

$$\int_{|x|}^{\infty}\frac{1}{\omega_n s^n}\,\frac{s^{n+1}e^{-s^2/2}}{2^{n/2}\Gamma(n/2+1)}\,ds=(2\pi)^{-n/2}e^{-|x|^2/2}.$$

The density of S gives $\mathbb{E}S^{-n}=2^{-n/2}/\Gamma(n/2+1)$. For $x>0$, integrating the gamma integral over $[x,x+1]$ gives $\Gamma(x+1)\geq x^x e^{-(x+1)}$. Monotonicity of L^p norms therefore gives

$$\mathbb{E}S^{-k}\leq(C_0/\sqrt{n})^k,\qquad 1\leq k\leq n,\qquad C_0:=e^{3/2}.$$

All coefficients in (3.1), viewed as a polynomial in σ^{-1}, are nonnegative. We may thus average term by term at $\sigma=S$ and replace each inverse moment of S by the corresponding power of $C_0/\sqrt{n}$. The mean value theorem now bounds the resulting difference of powers by $8C_0D\sqrt{n}\,\rho\exp(C_0(2D+1)\sqrt{n}\,\rho)$. □

In Hilbert–Schmidt orthonormal coordinates on the real space of Hermitian tuples, the GUE tuple has the law of $N^{-1/2}g$ for a standard Gaussian $g\in\mathbb{R}^{dN^2}$. Indeed, $\operatorname{Tr}(H_j^2)$ is the squared Euclidean norm of the coordinate vector of H_j. We use Gaussian concentration in the following form [CGP22, Proposition 4.4 of arXiv v1]. If F is K-Lipschitz in standard Gaussian coordinates, then

$$\mathbb{P}\{F-\mathbb{E}F\geq t\}\leq\exp(-t^2/(2K^2)),\qquad t>0. \tag{3.2}$$

Applying this to both F and $-F$ gives a two-sided estimate.

The bound $\mathbb{P}\{\|H_j^{(N)}\|\geq D+t\}\leq e^{-cNt}$ for $t>0$, from [CGP22, Proposition 2.11 of arXiv v1], gives a uniform bound M_0 on $\mathbb{E}\|H_j^{(N)}\|$ by integration. The map $g\mapsto\|H_j^{(N)}\|$ is $N^{-1/2}$-Lipschitz because the operator norm is bounded by the Hilbert–Schmidt norm. Using (3.2) and a union bound, we obtain

$$\mathbb{P}\{\max_j\|H_j^{(N)}\|>M\}\leq de^{-NM^2/8},\qquad M\geq 2M_0. \tag{3.3}$$

In particular the maximum has bounded moments of every fixed positive order, uniformly in N.

Proof of Theorem 1.3. In the same coordinates, define

$$q_N(x):=\left|\det T_N(N^{-1/2}x)\right|^2,\qquad Z_N:=q_N^{-1}(0).$$

The determinant has degree at most mN, so q_N is a nonnegative real polynomial of degree at most $2mN$, with $q_N(0)=1$.

Fix $M > 0$ and let $C_0, \alpha, \varepsilon_0 > 0$ be the constants of Proposition 2.1 with $A_0 = I_m$ and $A_j = R_j$. If Z_N is empty, the proposition excludes $s_{\min}(T_N(H^{(N)})) \le \varepsilon$ on $\max_j \|H_j^{(N)}\| \le M$ for $0 < \varepsilon \le \varepsilon_0$. Otherwise, on the same event the proposition gives

$$s_{\min}(T_N(H^{(N)})) \le \varepsilon \quad \Longrightarrow \quad \operatorname{dist}(g, Z_N) \le C_0\sqrt{N}\,\varepsilon^{\alpha}.$$

The factor $\sqrt{N}$ converts the Hilbert–Schmidt distance in the original matrix coordinates to distance in the standard Gaussian coordinates. Apply Lemma 3.1 with $D := 2mN$ and $\rho := C_0\sqrt{N}\,\varepsilon^{\alpha}$. Since $\sqrt{n} = \sqrt{d}\,N$, the resulting probability bound is

$$\mathbb{P}\{s_{\min}(T_N(H^{(N)})) \le \varepsilon,\ \max_j \|H_j^{(N)}\| \le M\} \le CN^{5/2}\varepsilon^{\alpha}\exp(CN^{5/2}\varepsilon^{\alpha}).$$

Take $M \ge 2M_0$ and choose $A > 0$ with $A\alpha > 7/2$. The right side is summable at $\varepsilon = N^{-A}$, as is the complementary norm event by (3.3). This proves (1.2). □

4. Normalized determinants of affine pencils

We prove Theorem 1.4. Write ν_N for the empirical spectral distribution of $L_N^*L_N$, with mass $1/(mN)$ per eigenvalue, and ν for the distribution of L^*L under τ_m. Thus $u = \frac{1}{2}\int \log x\, d\nu(x)$. If all the random coefficients vanish, then $U_N = u$ and $\Delta_{mN}(L_N) = e^u$ identically, and (1.5) in the finite case follows by choosing $\varepsilon_0 < s_{\min}(A_0)$. We assume below that some A_j with $j \ge 1$ is nonzero.

The normalized trace moments of $L_N^*L_N$ converge in probability to those of L^*L. Indeed, expansion of each matrix-valued moment gives a finite linear combination of scalar trace moments with fixed coefficients. Since $\|L_N\| \le C_A(1 + \max_j \|H_j^{(N)}\|)$ for a fixed constant C_A, the measures are supported in a common compact interval with probability tending to one by (3.3). Polynomial approximation gives $\nu_N \Rightarrow \nu$ in probability. The expected measures $\mathbb{E}\nu_N$ also converge weakly to ν, because bounded test integrals converging in probability converge in mean.

4.1. **Generic invertibility.** Assume $u > -\infty$, so $\nu(\{0\}) = 0$. Let $\mathcal{X} := \operatorname{span}_{\mathbb{C}}\{A_0, \dots, A_d\}$. The square tensor blow-up of $\mathcal{X}$ of size N is $\mathcal{X} \otimes M_N(\mathbb{C})$, viewed as a subspace of $M_{mN}(\mathbb{C})$. The maximum rank of a matrix in this space is a multiple of N, by the rank regularity lemma of Ivanyos, Qiao and Subrahmanyam in [DM17, Definitions 2.1–2.2 and Proposition 2.3]. This maximum equals the maximum r_N of $\operatorname{rank}(A_0 \otimes I_N + \sum_j A_j \otimes X_j)$ over complex matrices $X_1, \dots, X_d$ of size N. Indeed, a tuple maximizing the rank of $\sum_{j=0}^{d} A_j \otimes X_j$ can be chosen with X_0 invertible, since these two nonempty Zariski open conditions intersect. Right multiplication by $I_m \otimes X_0^{-1}$ preserves rank and yields an affine tuple. The reverse inequality follows by setting $X_0 = I_N$.

A real basis of the Hermitian matrices is also a complex basis of $M_N(\mathbb{C})$. Consequently, a nonzero polynomial in the complex matrix entries remains a nonzero polynomial on the real Hermitian coordinates, by the polynomial identity principle. If $r_N > 0$, an r_N-minor that is a nonzero polynomial therefore vanishes with probability zero on the Gaussian tuple, so L_N has rank r_N almost surely. The same rank assertion is immediate for $r_N = 0$. Whenever $r_N < mN$, divisibility by N gives $\mathbb{E}\nu_N(\{0\}) \ge 1/m$. If this occurred along an infinite subsequence, the closed-set bound for weak convergence of $\mathbb{E}\nu_N$ would give $\nu(\{0\}) \ge 1/m$. Thus $r_N = mN$ for all sufficiently large N, and the determinant polynomial on Hermitian inputs is then nonzero.

4.2. **A norm cutoff depending on the singular value.** We continue to assume $u > -\infty$. The maps $F_{k,\ell}$ from the proof of Proposition 2.1 form a fixed finite family. By (2.4), a residual at most ε on $\max_j \|H_j\| \le M$ gives $|x|^2 \le C(1 + M^2)$ and $|F_{k,\ell}(x)| \le \varepsilon$. For the maps with nonempty zero set, the distance inequality and the Hermitian lift in that proof give a perturbation of size at most $C(1 + M^2)\varepsilon^a$, where $a > 0$ is the minimum of their distance exponents. If there are no such maps, take $a := 1$.

For a map F on $\mathbb{R}^p$ with empty zero set, the map

$$(x, y) \longmapsto (yF(x), y(y-1))$$

has zero set $\mathbb{R}^p \times \{0\}$. It has degree at most three, and its value at $(x, 1)$ has norm $|F(x)|$. The polynomial distance inequality used in Proposition 2.1, applied at distance one from this zero set, gives

$$|F(x)| \geq c_F(2 + |x|^2)^{-E_F}, \qquad E_F := 3 \cdot 15^p.$$

Let E_* be the largest of these exponents, or 1 if all the zero sets are nonempty. Choose $0 < \theta \leq \min(a/2, 1/(2E_*))$ and take $M := \varepsilon^{-\theta/2}$. For an empty zero type the residual would be at least $c'_F \varepsilon^{\theta E_F} \geq c'_F \varepsilon^{1/2}$, which exceeds ε for all sufficiently small ε. Every remaining type gives a Hermitian perturbation of size at most $C\varepsilon^b$ for $b := a - \theta > 0$.

In standard Gaussian coordinates let Z_N be the zero set of $|\det L_N(N^{-1/2}g)|^2$, a nonzero polynomial of degree at most $2mN$ by the preceding rank argument. On $\max_j \|H_j^{(N)}\| \leq M$, a least singular value at most ε puts g within $C\sqrt{N}\,\varepsilon^b$ of Z_N. If Z_N is empty, that event is empty by the construction of the perturbation. With $n = dN^2$, Lemma 3.1 and (3.3) give

$$\mathbb{P}\{s_{\min}(L_N) \leq \varepsilon\} \leq CN^{5/2}\varepsilon^b e^{CN^{5/2}\varepsilon^b} + d\exp(-N\varepsilon^{-\theta}/8)$$

for all sufficiently large N and sufficiently small ε. The second term is bounded by a constant times ε^b, and the first exponential factor is bounded when $N^{5/2}\varepsilon^b \leq 1$. This proves (1.5). Taking B with $Bb > 7/2$ and applying the first Borel–Cantelli lemma gives $s_{\min}(L_N) \geq N^{-B}$ eventually almost surely.

4.3. **Singular-value averages.** For finite u, we first bound the expected mass of small squared singular values. Write G_N and G for the Cauchy transforms of ν_N and ν, respectively, with kernel $(w - x)^{-1}$. The matrix-valued resolvent comparison of [CGP22, Corollary 1.3 of arXiv v1] applies to $L_N^* L_N$ with fixed amplification size m and deterministic matrices $A_0, \ldots, A_d$ and their adjoints. Interchanging the two tensor factors identifies the limiting transform with G under the normalized trace τ_m. For $\Im w > 0$ it gives

$$|\mathbb{E}G_N(w) - G(w)| \leq CN^{-2}((\Im w)^{-5} + (\Im w)^{-2}).$$

By [SS15, Theorem 1.1(5)–(6)], G has a convergent expansion in fractional powers at zero. The bounds $|G(\mathrm{i}t)| \leq 1/t$ and $\mathrm{i}tG(\mathrm{i}t) \to \nu(\{0\}) = 0$ exclude all powers at most -1 on the positive imaginary axis. Thus $t|\Im G(\mathrm{i}t)| \leq Ct^\beta$ for some $0 < \beta \leq 1$ and $0 < t \leq 1$. The inequality $\mathbf{1}_{[0,t]}(x) \leq 2t^2/(x^2 + t^2)$ for $x \geq 0$ yields

$$\mathbb{E}\nu_N([0, t]) \leq 2\mathbb{E}C_N(t) \leq Ct^\beta, \qquad t_N := N^{-\gamma} \leq t \leq 1, \quad 0 < \gamma < \frac{2}{4 + \beta}, \tag{4.1}$$

where

$$C_N(t) := \frac{1}{mN} \sum_i \frac{t^2}{s_i(L_N)^4 + t^2} = -t\Im G_N(\mathrm{i}t).$$

Here the comparison error $CN^{-2}t^{-4}$ is absorbed into Ct^β for $t \geq t_N$.

For the almost-sure estimates, let $f : [0, \infty) \to \mathbb{R}$ be Lipschitz and put

$$\Phi_{N,f}(g) := \frac{1}{mN} \sum_i f(s_i(L_N(g))),$$

where $L_N(g)$ denotes evaluation at the tuple with coordinates $N^{-1/2}g$. For matrices X, Y of the same size, with singular values ordered decreasingly, the Frobenius singular-value inequality is

$$\sum_i (s_i(X) - s_i(Y))^2 \leq \|X - Y\|_{\mathrm{HS}}^2 \qquad \text{[Mec19, Corollary 4.39]}.$$

Together with Cauchy–Schwarz, it bounds the change of the normalized average by $\mathrm{Lip}(f)\|X - Y\|_{\mathrm{HS}}/\sqrt{mN}$. Writing $\kappa_A := (\sum_{j=1}^{d} \|A_j\|_{\mathrm{HS}}^2)^{1/2}$, the tensor Hilbert–Schmidt identity gives

$$\|L_N(g) - L_N(g')\|_{\mathrm{HS}} \le \kappa_A N^{-1/2} |g - g'|.$$

Consequently (3.2) gives, for every $r > 0$,

$$\mathbb{P}\{|\Phi_{N,f} - \mathbb{E}\Phi_{N,f}| > r\} \le 2\exp\left(-\frac{mN^2 r^2}{2\kappa_A^2 \,\mathrm{Lip}(f)^2}\right) \tag{4.2}$$

when f is nonconstant.

4.4. **Almost-sure convergence.** First suppose $u = -\infty$. For fixed $\delta > 0$, the regularization

$$U_{N,\delta} := \frac{1}{2}\int \log(x + \delta^2)\, d\nu_N(x)$$

has a fixed lower bound $\log \delta$ and a uniformly integrable positive part by the norm moments from (3.3). Weak convergence in probability and truncation therefore give convergence in L^1 to $u_\delta := \frac{1}{2}\int \log(x+\delta^2)\, d\nu(x)$. The function $s \mapsto \frac{1}{2}\log(s^2 + \delta^2)$ is $(2\delta)^{-1}$-Lipschitz. Hence (4.2) is summable over N for each fixed deviation, and Borel–Cantelli for deviations $1/k$, $k \ge 1$, gives $U_{N,\delta} \to u_\delta$ almost surely. Intersect the resulting events over a sequence $\delta \downarrow 0$. Since $U_N \le U_{N,\delta}$ and $u_\delta \downarrow -\infty$, this gives $U_N \to -\infty$ almost surely.

Suppose now that $u > -\infty$ and use $t_N = N^{-\gamma}$ from (4.1). The function $s \mapsto t^2/(s^4 + t^2)$ is $4t^{-1/2}$-Lipschitz, as is seen by rescaling $s = \sqrt{t}\, y$. At $t = t_N$, (4.2) bounds the probability of a deviation larger than $N^{-\gamma\beta}$ for $C_N(t_N)$ by

$$2\exp(-cN^{2-\gamma(1+2\beta)}).$$

The exponent is positive because $\beta \le 1$ and $\gamma(4+\beta) < 2$. By Borel–Cantelli and (4.1), $\nu_N([0, t_N]) \le CN^{-\gamma\beta}$ eventually almost surely.

Put $V_N(t) := \frac{1}{2}\int \log\max(x, t)\, d\nu_N(x)$. The function $s \mapsto \log\max(s, \sqrt{t})$ is $t^{-1/2}$-Lipschitz, so (4.2) and Borel–Cantelli also give $V_N(t_N) - \mathbb{E}V_N(t_N) \to 0$ almost surely. For fixed $0 < \eta < 1$, weak convergence and the uniform norm moments give

$$\mathbb{E}V_N(\eta) \longrightarrow v(\eta) := \frac{1}{2}\int \log\max(x, \eta)\, d\nu(x).$$

By layer integration and (4.1), for $t_N < \eta$,

$$0 \le \mathbb{E}V_N(\eta) - \mathbb{E}V_N(t_N) \le \frac{1}{2}\int_{t_N}^{\eta} \mathbb{E}\nu_N([0,t])\,\frac{dt}{t} \le C\eta^\beta.$$

Since $v(\eta) \downarrow u$, this proves $\mathbb{E}V_N(t_N) \to u$. On the eventual singular-value floor from (1.5),

$$0 \le V_N(t_N) - U_N \le B(\log N)\nu_N([0, t_N]) \longrightarrow 0 \qquad \text{almost surely.}$$

This proves $U_N \to u$ almost surely in the finite case as well. All Borel–Cantelli arguments used summable probabilities of individual events, so they apply to every coupling across sizes.

4.5. **Convergence of expectations.** For finite u, choose B with $Bb > 5/2$ and $2B > \gamma$, and put $a_N := N^{-2B}$. For $0 < t \le a_N$, (1.5) gives $\mathbb{E}\nu_N([0,t]) \le CN^{5/2}t^{b/2}$, since positive mass requires a singular value at most $\sqrt{t}$. Integrating with respect to dt/t up to a_N gives a bound $CN^{5/2-Bb}$. Between a_N and t_N, monotonicity and (4.1) give a bound $C(\log N)N^{-\gamma\beta}$. On $[t_N, \eta]$, (4.1) bounds the integral by $C\eta^\beta$ for fixed $0 < \eta < 1$ and all large N. Together with the endpoint term $|\log\eta|\mathbb{E}\nu_N([0,\eta])$, the layer integral bounds the expected negative logarithmic tail by

$$\mathbb{E}\int_{(0,\eta)} |\log x|\, d\nu_N(x) \le C\eta^\beta(1 + |\log\eta|) + C(\log N)N^{-\gamma\beta} + CN^{5/2-Bb}.$$

Generic invertibility holds at all sufficiently large sizes, so the finite bound also gives integrability of the negative logarithm at these sizes. The norm moments from (3.3) control the positive tail uniformly, since its integral above $R > 1$ is bounded by $2\log^+ \|L_N\|\mathbf{1}_{\{\|L_N\|>\sqrt{R}\}}$. The integrals of the logarithm with fixed upper and lower cutoffs converge in L^1 by boundedness and weak convergence in probability. Removing the cutoffs using the two tail estimates and the integrability of $\log x$ under ν proves $\mathbb{E}|U_N - u| \to 0$.

If $u = -\infty$, then $U_N^+ \le \log^+ \|L_N\|$ has uniformly bounded expectation. For each $\delta > 0$, the L^1 convergence of the regularized logarithm gives $\limsup_N \mathbb{E}U_N \le u_\delta$. Letting $\delta \downarrow 0$ yields $\mathbb{E}U_N \to -\infty$.

Finally $\Delta_{mN}(L_N) = e^{U_N}$ converges almost surely to e^u in both cases. Since $\Delta_{mN}(L_N) \le \|L_N\|$, the uniform norm moments of order greater than any given $p > 0$ imply $\mathbb{E}|\Delta_{mN}(L_N) - e^u|^p \to 0$.

5. From the pencil estimate to Brown measure

5.1. **Linearization.** We prove Theorem 1.2. For constant P, both measures in the theorem are the same point mass. Assume henceforth that P is nonconstant, and set $A_N := P(H^{(N)})$ and $a := P(s)$.

Fix $z \ne P(0)$. We represent $P - z$ as a Schur complement of an affine pencil, using one block for each word of length at least two (compare [Han26, Section 2.2]). For a term $c_w w$ with $w = x_{i_1} \cdots x_{i_k}$ and $k \ge 2$, let V_w be the $(k-1) \times (k-1)$ matrix with entries $(V_w)_{r,r+1} := x_{i_{r+1}}$ for $1 \le r \le k-2$ and zeros elsewhere. Set

$$Q_w := I_{k-1} - V_w, \qquad B_w := (c_w x_{i_1}, 0, \dots, 0), \qquad C_w := (0, \dots, 0, -x_{i_k})^{\mathsf{T}}.$$

The polynomial inverse $Q_w^{-1} = I_{k-1} + V_w + \cdots + V_w^{k-2}$ gives $-B_w Q_w^{-1} C_w = c_w w$. Let Q be the block diagonal matrix with blocks Q_w, and assemble the B_w as a block row B and the C_w as a block column C in the same order. With a_z the affine part of $P - z$, the pencil

$$L_z(x) := \begin{pmatrix} a_z & B \\ C & Q \end{pmatrix}$$

has Schur complement $a_z - BQ^{-1}C = P - z$. If P is affine we simply take $L_z := P - z$. When present, the auxiliary block Q evaluates to a unit upper triangular matrix with determinant one. In either case we obtain

$$\det L_z(H^{(N)}) = \det(A_N - zI_N). \tag{5.1}$$

Write q for the size of L_z and let $D_z := L_z(0) = \operatorname{diag}(P(0) - z, 1, \dots, 1)$. Left multiplication by $D_z^{-1} \otimes I_N$ gives a monic pencil

$$\widetilde{T}_N(z) := (D_z^{-1} \otimes I_N) L_z(H^{(N)}) = I_q \otimes I_N + \sum_{j=1}^{d} \widetilde{R}_j(z) \otimes H_j^{(N)}.$$

The upper left $N \times N$ block of its inverse, when it exists, is $(P(0) - z)(A_N - zI_N)^{-1}$. Taking operator norms yields

$$s_{\min}(A_N - zI_N) \ge |P(0) - z|\, s_{\min}(\widetilde{T}_N(z)). \tag{5.2}$$

This also holds in the singular case because $\widetilde{T}_N(z)$ is invertible exactly when $A_N - zI_N$ is invertible. For the fixed shift z, Theorem 1.3 and (5.2) give a $B_z > 0$ such that

$$\mathbb{P}\{s_{\min}(A_N - zI_N) < N^{-B_z}\} \longrightarrow 0. \tag{5.3}$$

For each N, $|\det(A_N - zI_N)|^2$ is a real polynomial in the Gaussian coordinates and is nonzero at the zero input tuple. Its zero set has Lebesgue measure zero, and thus probability zero.

5.2. **The logarithmic tail.** For each fixed $z \neq P(0)$, set $Q_z := (P-z)^*(P-z)$. Let ν_N^z be the empirical law of the squared singular values of $A_N - zI_N$, and let ν^z be the spectral distribution of $Q_z(s)$ with respect to τ. Write

$$G_N(w) := \int \frac{1}{w-x}\, d\nu_N^z(x), \qquad G(w) := \int \frac{1}{w-x}\, d\nu^z(x).$$

The scalar polynomial Q_z is self-adjoint and nonconstant. Indeed, if P_k is the top homogeneous part of P, then the top homogeneous part of Q_z is $P_k^* P_k$. For words u, v of length k, each word u^*v determines both u and v, so a word u with nonzero coefficient in P_k contributes a nonzero coefficient to u^*u.

The resolvent comparison of [CGP22, Corollary 1.3 of arXiv v1], applied to Q_z with scalar coefficients, gives

$$|\mathbb{E}G_N(w) - G(w)| \leq C_z N^{-2}((\Im w)^{-5} + (\Im w)^{-2}), \qquad \Im w > 0. \tag{5.4}$$

By [SS15, Theorem 1.1(5)–(6) and Corollary 1.2], G has a convergent expansion in fractional powers at zero and ν^z has no atoms. The bound $|G(\mathrm{i}t)| \leq t^{-1}$ excludes powers below -1 along the positive imaginary axis. Also,

$$\mathrm{i}tG(\mathrm{i}t) = \int \frac{\mathrm{i}t}{\mathrm{i}t - x}\, d\nu^z(x) \longrightarrow \nu^z(\{0\}) = 0.$$

Thus the expansion has no term of order -1 either, and for some $0 < \beta_z \leq 1$,

$$t|\Im G(\mathrm{i}t)| \leq C_z t^{\beta_z}, \qquad 0 < t \leq 1.$$

Using $\mathbf{1}_{[0,t]}(x) \leq 2t^2/(x^2+t^2)$ for $x \geq 0$ and (5.4), we obtain

$$\mathbb{E}\nu_N^z([0,t]) \leq C_z(t^{\beta_z} + N^{-2}t^{-4}), \qquad 0 < t \leq 1.$$

Choose $0 < \gamma_z < 2/(4+\beta_z)$ and write $t_N := N^{-\gamma_z}$. The resolvent error can then be absorbed into the first term for $t \geq t_N$, giving

$$\mathbb{E}\nu_N^z([0,t]) \leq C_z t^{\beta_z}, \qquad t_N \leq t \leq 1. \tag{5.5}$$

The same kernel inequality gives $\nu^z([0,t]) \leq C_z t^{\beta_z}$ for all $0 < t \leq 1$.

On the event $s_{\min}(A_N - zI_N) \geq N^{-B_z}$, the contribution to the negative logarithm from $0 < x \leq t_N$ is at most $2B_z(\log N)\nu_N^z([0,t_N])$. The expectation of this bound tends to zero by (5.5). For $t_N < x < \varepsilon < 1$, write $-\log x = |\log \varepsilon| + \int_x^\varepsilon dt/t$. Taking expectations and applying (5.5) gives

$$\mathbb{E}\int_{(t_N,\varepsilon)} |\log x|\, d\nu_N^z(x) \leq C_z\varepsilon^{\beta_z}|\log\varepsilon| + C_z\int_{t_N}^{\varepsilon} t^{\beta_z - 1}\, dt \leq C_z'\varepsilon^{\beta_z}(1 + |\log\varepsilon|)$$

for all sufficiently large N at each fixed ε. Markov's inequality and (5.3) now give, for every $\delta > 0$,

$$\lim_{\varepsilon \downarrow 0} \limsup_{N\to\infty} \mathbb{P}\left\{\int_{(0,\varepsilon)} |\log x|\, d\nu_N^z(x) > \delta\right\} = 0. \tag{5.6}$$

The bound on $\nu^z([0,t])$ gives a vanishing logarithmic tail for ν^z by the same integration formula.

5.3. **Convergence of potentials.** The GUE moment convergence and norm bounds from [CGP22, pp. 1–2 of arXiv v1] give weak convergence of ν_N^z to ν^z in probability. In fact, normalized trace moments converge, and both measures are supported in a common fixed compact interval with probability tending to one. Polynomial approximation on that interval gives convergence against bounded continuous functions. Consequently we may first truncate the logarithm to a bounded continuous function, with a fixed upper cutoff above that interval. Then (5.6) and the limiting tail bound let us send the lower cutoff to zero. We conclude that, for every fixed $z \neq P(0)$,

$$u_N(z) := \frac{1}{N}\log|\det(A_N - zI_N)| = \frac{1}{2}\int \log x\, d\nu_N^z(x) \xrightarrow{\mathbb{P}} u_a(z) = \tau(\log|a - z1_{\mathcal{M}}|). \tag{5.7}$$

5.4. **Almost-sure weak convergence.** For each fixed $z \neq P(0)$, apply Theorem 1.4 to the unnormalized pencil L_z of size q. Equation (5.1) identifies its normalized logarithmic determinant with $u_N(z)/q$. The theorem gives the almost-sure limit of this quantity, which by uniqueness of probability limits and (5.7) equals $u_a(z)/q$. In particular,

$$u_N(z) \longrightarrow u_a(z) \quad \text{almost surely for each fixed } z \neq P(0).$$

The potentials $u_N(\omega, z)$ are jointly measurable, since they are logarithms of absolute values of determinants with measurable matrix entries and continuous dependence on z. Fubini's theorem on disks of integer radii gives one event of probability one on which $u_N(z) \to u_a(z)$ for planar almost every z. By (3.3) and the first Borel–Cantelli lemma, the input norms are eventually bounded by a fixed constant on another event of probability one. On the intersection of these two events, the norm bound on the inputs implies that all sufficiently large N have $\|A_N\| \leq M$ for a fixed deterministic M which we take larger than $\|a\|$.

Fix a sample point in this intersection and a compact $K \subset \mathbb{C}$. The functions $z \mapsto \log|z - w|$ have uniformly bounded $L^2(K)$ norm for $|w| \leq M$, since the translated sets $K - w$ lie in one bounded disk and $\log|\cdot|$ is locally square integrable. Using the determinant formula $u_N(z) = N^{-1}\sum_{k=1}^{N} \log|z - \lambda_k(A_N)|$ and Jensen's inequality gives $\|u_N\|_{L^2(K)} \leq C_{M,K}$ for all sufficiently large N. The potential representation of the Brown measure gives $\|u_a\|_{L^2(K)} \leq C_{M,K}$ as well.

For $\delta > 0$, let $E_{N,\delta} := \{z \in K : |u_N(z) - u_a(z)| > \delta\}$, taking the potentials up to changes on planar null sets. Dominated convergence of indicators gives $|E_{N,\delta}| \to 0$, where $|\cdot|$ denotes planar Lebesgue measure. For all sufficiently large N, Cauchy–Schwarz yields

$$\|u_N - u_a\|_{L^1(K)} \leq \delta|K| + 2C_{M,K}|E_{N,\delta}|^{1/2}.$$

Letting $N \to \infty$ and then $\delta \downarrow 0$ proves $u_N \to u_a$ in local L^1 at this sample point.

For $\varphi \in C_c^\infty(\mathbb{C})$, the distributional identity $\Delta \log|z| = 2\pi\delta_0$ and Definition 1.1 give

$$\int \varphi \, d\mu_{A_N} = \frac{1}{2\pi}\int_{\mathbb{C}} u_N \Delta\varphi \, d^2z \;\longrightarrow\; \frac{1}{2\pi}\int_{\mathbb{C}} u_a \Delta\varphi \, d^2z = \int \varphi \, d\mu_a^{\mathrm{Br}}.$$

Both probability measures are supported in the disk of radius M for all sufficiently large N. Smooth compactly supported functions approximate any bounded continuous test function uniformly on this disk. The local L^1 conclusion holds on the same event for every compact set, so the resulting weak convergence holds against all bounded continuous real-valued test functions on that event. This proves Theorem 1.2.

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